Solution Requirements

Date	15 March 2026
Team ID	SWTID-2026-5240
Project Name	Streamlining IT Procurement: Automating Standard Laptop Procurement
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

- Date: 15 March 2026
- Team ID: SWTID-2026-5240
- Project Name: Automated Sketch-to-Website Generation using OCR & Flask
- Maximum Marks: 4 Marks

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority (High/Medium/Low)
1	Authentication	Web UI meedha user role Based access, Flask session tokens, mariyu secure input file validation untundhi.	Medium
2	Authorization levels	User level (sketch upload, code output preview download) mariyu Admin level (model parameters update, contour detection settings manage cheyadam).	Medium
3	External interfaces	OpenCV computer vision libraries, Tesseract/EasyOCR engine, mariyu Flask backend APIs connect avuthayi.	High
4	Transactions processing	Sketch image processing request upload chesina ventane image conversion, contour mapping, mariyu HTML/CSS/JS code creation fast-ga jaruguthundhi.	High
5	Reporting	Processed sketch extraction logs, detected layout JSON structures, mariyu web page preview reports system generate chesthundhi.	Medium
6	Business rules, etc.	Hand-drawn wireframe images lo elements (buttons, inputs, headings) 80%+ clear boundaries tho undali. Unknown shapes ki default blocks allocate avuthayi.	High
7	Compliance to laws or regulations	User upload chesina sketch assets mariyu source code files privacy-ki OWASP top-10 web security standards follow avuthayi.	Medium
8	Other	Generated HTML5/CSS3 output standard web browsers (Chrome, Edge, Firefox) lo	High

		mobile-responsive layout ga render avuthundhi.	

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
1	Performance & Speed	Processing speed for wireframe parsing and code generation.	Image upload ayyaka code preview generation < 3 seconds lo poorthi avuthundhi.
2	Scalability	Multiple image processing requests run chese capability.	System okey sari up to 100 concurrent wireframe extraction jobs support chesthundhi.
3	Security & Data Privacy	Uploaded images and generated code security.	Data transit lo HTTPS encryption mariyu session-based file cleanup setup untundhi.
4	Reliability & Availability	Continuous application processing uptime.	Local/Cloud Flask application server 99.5% uptime maintain chesthundhi.
5	Usability & Accessibility	Simple drag-and-drop wireframe image upload user interface.	W3C Web Standards compliance mariyu intuitive clean UI/UX layout.
6	Other (Maintainability)	Modular code architecture for future expansion.	Flask modular routes, Python clean code standards, and standard CSS component styling.